The Infinite Loop Considered Harmless

The lambda calculus derived from itself, over one symbol

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October 2026

The bent card. The accomplice marks one card so the mark believes the quarter can be followed; this is that card, and the throw unbends it.

Reading note

The previous card said that the formal grammar of the construction has one terminal and that reading that terminal as a successor gives arithmetic. This card writes the calculus in which both facts are theorems, derives the calculus from itself in the standard way, exhibits the term that never stops, and proves that it is harmless in a precise sense: its record is the one-symbol tape, its reading is the counter, and it writes no quarter. Grades are as in the companion card: *proved here*, *standard*, *proposed*. In the game this is the bent card: the one with a corner turned up so the mark thinks he can follow the quarter under it. The bend is a receipt supplied from outside, not a reading of the card, and the throw unbends it. Here the bend is the count of demonstrations, the five cards read as a stack; Proposition 3.3 is the throw.

1 The calculus from one symbol

Definition 1.1 (Terms)**.** Terms are generated by three rules: a variable x is a term; if M is a term and x a variable then $\lambda x.M$ is a term (abstraction); if M and N are terms then MN is a term (application). The one computation rule is β: $(\lambda x.M)N \to M[x := N]$ [2, 1]. *Standard.*

Definition 1.2 (Numerals)**.** For $n \geq 0$, $\overline{n} = \lambda f.\lambda x.\, f^n x$, where $f^n x$ is f applied n times to x. *Standard.*

Proposition 1.3 (The one-terminal grammar is the numerals)**.** *Let the grammar of the companion card have terminal* T *and let* T *be read as application of* f*. Then every sentence of the grammar whose nonterminals are all erased is* $f^n x$ *for the number* n *of terminals it contains, and the reader that counts terminals is the map* $\overline{n} \mapsto n$ *obtained by applying* $\overline{n}$ *to the successor and zero.* Proved here.

Proof. Erasing nonterminals leaves a string of n copies of T; reading each as one application of f gives $f^n x$. Applying $\overline{n}$ to succ and 0 gives $\mathrm{succ}^n 0 = n$. □

The abstraction λf is the grammar requiring a name for its symbol. The application that supplies succ is the choice. The numeral says nothing about what f is; that is handed in.

Proposition 1.4 (Two readings, two arithmetics)**.** *Addition of numerals is* $\lambda m.\lambda n.\lambda f.\lambda x.\, mf(nfx)$ *and multiplication is* $\lambda m.\lambda n.\lambda f.\, m(nf)$*. Reading* T *by tally uses only the first; reading a string of* n *terminals positionally in base two, as* $2^n - 1$*, uses the second.* Standard *for the definitions; the identification of the two readings with the companion card's tally and register is* proved here *by Proposition 1.3.*

2 The calculus from itself

A term can be written as a term, and a term can evaluate that term. This is the self-interpreter, and the construction below is Mogensen's [4]. *Standard; reproduced here.*

Definition 2.1 (Encoding)**.**

$$\ulcorner x \urcorner = \lambda a.\lambda b.\lambda c.\, a\, x, \qquad \ulcorner MN \urcorner = \lambda a.\lambda b.\lambda c.\, b \ulcorner M \urcorner \ulcorner N \urcorner, \qquad \ulcorner \lambda x.M \urcorner = \lambda a.\lambda b.\lambda c.\, c\, (\lambda x.\ulcorner M \urcorner).$$

Definition 2.2 (Fixed-point combinator)**.** $\mathbf{Y} = \lambda f.(\lambda x.f(xx))(\lambda x.f(xx))$, so that $\mathbf{Y}f \to f(\mathbf{Y}f)$ for every f.

Theorem 2.3 (Self-interpretation)**.** *Let*

$$\mathbf{E} = \mathbf{Y}\,(\lambda e.\lambda m.\, m\,(\lambda x.x)\,(\lambda p.\lambda q.\,(e\,p)(e\,q))\,(\lambda p.\lambda v.\, e\,(p\,v))).$$

Then $\mathbf{E}\ulcorner M\urcorner =_\beta M$ *for every term* M*.* Standard *[4].*

Proof. Induction on M. For a variable, $\mathbf{E}\ulcorner x\urcorner \to \ulcorner x\urcorner(\lambda x.x)(\dots)(\dots) \to (\lambda x.x)x \to x$. For an application, $\mathbf{E}\ulcorner MN\urcorner \to (\mathbf{E}\ulcorner M\urcorner)(\mathbf{E}\ulcorner N\urcorner) =_\beta MN$ by the hypothesis. For an abstraction, $\mathbf{E}\ulcorner \lambda x.M\urcorner \to \lambda v.\mathbf{E}(\ulcorner M\urcorner[x := v])$, and $\ulcorner M\urcorner[x := v] = \ulcorner M[x := v]\urcorner$ since the encoding of a variable is the variable itself under a; the hypothesis gives $\lambda v.M[x := v]$, which is $\lambda x.M$ up to renaming. □

This is what "the calculus derived from itself" means: the evaluator is a term of the calculus, the syntax is a term of the calculus, and the theorem that one applied to the other returns the original is a theorem of the calculus. Nothing outside it was used. The one thing it does not do is choose f: the self-interpreter evaluates numerals faithfully and leaves their abstraction abstract.

3 The loop

Definition 3.1 (The term that never stops)**.** $\mathbf{\Omega} = (\lambda x.xx)(\lambda x.xx)$. Then $\mathbf{\Omega} \to \mathbf{\Omega}$, and $\mathbf{\Omega}$ has no normal form [1]. *Standard.*

The self-interpreter inherits it: $\mathbf{E}\ulcorner\mathbf{\Omega}\urcorner$ reduces forever, and $\mathbf{Y}$ itself is built from the same self-application. Dijkstra's objection to the unrestricted jump [3] was that it severs the record of a computation from the text that describes it: after a jump, where the process is cannot be read from where the program is. The loop is the jump that jumps to itself. The title claims it is harmless, and the claim has three parts.

Proposition 3.2 (The loop's record is the one-symbol tape)**.** *Read each reduction step as writing one terminal. Then the record of* $\mathbf{\Omega}$ *after* n *steps is the string* T^n*, the tally reads* n*, and the positional reading is* $2^n - 1$*. No step writes a different symbol, and no prefix of the record determines any later step differently from the first.* Proved here.

Proof. Each step is the same β-redex reducing to itself; by Proposition 1.3 the step record is one terminal per step. The last clause is the statement that $\mathbf{\Omega} \to \mathbf{\Omega}$ is the only step. □

So the loop is the counter and nothing else: a record that grows by one unchanged symbol per step, whose positional reading grows exponentially while the thing read never changes. That is the de Sitter counter of the numerical card, exactly, and it is harmless for the same reason: it changes nothing but the count.

Proposition 3.3 (The loop writes no quarter)**.** *No prefix of the record of* $\mathbf{\Omega}$ *determines a value of any free variable, and* $\mathbf{\Omega}$ *has none. In particular the loop supplies no* f *to any numeral and decides no comparison.* Proved here.

Proof. $\mathbf{\Omega}$ is closed, and its reduction sequence contains only $\mathbf{\Omega}$; a record consisting of one closed term repeated determines nothing about any variable. □

Proposition 3.4 (Harm is a loop read as a value)**.** *Let a record system accept a term as the value of a computation only when the computation has reached a normal form. Then* $\mathbf{\Omega}$ *is never accepted as a value, and a system that accepted it would accept every term, since for every* M *the term* $(\lambda x.M)\mathbf{\Omega}$ *reduces to* M *along one path and loops along another.* Proved here.

Proof. The first clause is the definition. For the second, if a looping reduction counted as reaching a value, then the looping path of $(\lambda x.M)\mathbf{\Omega}$ would certify a value for a term whose other path certifies M, for arbitrary M: every term would be a value of every other. □

That is the only harm the loop can do, and it is done by the reader, not by the loop. The loop considered as a record is a counter. The loop considered as a value is a contradiction. A record system stays consistent by keeping the arm that says "not yet" separate from the arm that says "done": the computation that has not halted is absence of evidence, and reading it as evidence of absence is the one mistake. The typed calculus in which the middle card is written makes this structural: self-application is not well typed there, $\mathbf{Y}$ has no type, every term normalizes, and the loop cannot be written at all. The price is that fixed points must be handed in from outside, which is the quarter in its λ-calculus form: $\mathbf{Y}f = f(\mathbf{Y}f)$ supplies a fixed point by a combinator in the untyped calculus and by a choice in the typed one.

4 What is proposed

Remark (The counter is the loop)*.* The companion cards carry a counter that increases by exactly one per written entry, a register that is the positional reading of a one-symbol tape, and a verdict decided once at the top. Propositions 3.2 and 3.3 identify the counter with the record of $\mathbf{\Omega}$ read one terminal per step: the publication's growth is the loop's growth, harmless because the loop's symbol never changes. The further identification of the loop with the global time of the de Sitter slicing, and of its exponential positional reading with the scale factor, is *proposed*, as it was in the numerical card. What would discharge it is unchanged: the cosmological constant as a residue is derived; the map from the record's steps to a physical clock is not.

Remark (Considered harmless)*.* Dijkstra was right about the jump, and the loop is the jump that is harmless, for one reason: it never leaves the text. Where the process is and where the program is coincide at every step, because the step is the program. A loop that wrote a second symbol would be a computation; a loop that wrote the first symbol forever is a clock; and a clock is the one thing a record cannot be wrong about, since it asserts nothing but that another entry was written. The harm begins at the first reader who takes the clock for a value. The middle card's press check reads once, at the top, and decides exclusive-or between two readings; it never reads the loop, because the loop has no top.

References

[1] Hendrik Pieter Barendregt. *The Lambda Calculus: Its Syntax and Semantics*. North-Holland, Amsterdam, revised edition, 1984.

[2] Alonzo Church. An unsolvable problem of elementary number theory. *American Journal of Mathematics*, 58(2):345–363, 1936.

[3] Edsger W. Dijkstra. Letters to the editor: go to statement considered harmful. *Communications of the ACM*, 11(3):147–148, 1968.

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